6GGW / NetSwitch — installation, hardware/software & compatibility

Everything needed to stand the system up on a Linux (RHEL/CentOS/Rocky/Alma/Fedora/ Gentoo/Debian/Ubuntu) or Windows Server (2016–2025 Datacenter) box, plus the compatibility matrix, ports/sockets, patch guidance, and the hotlink-protection URL-rewrite rules you asked for.

> **MFLOPS.** Every “MFLOPS HERE NEEDED” is filled by the **report engine** (`ggw_report`). Run it on the target box and it writes a report with that box’s real double-precision MFLOPS (dense matmul + sustained FMA), CPU/RAM/disk, and an OS compatibility verdict. Reference figure from my build box (AMD EPYC-Genoa, one thread, `-O2`): **~3,400 MFLOPS matmul / ~7,200 MFLOPS FMA**. Your hardware will print its own — that is the number to quote per machine.

Hardware requirements

Tier	CPU	RAM	Disk	Use
Thin client (phone/PC)	2 cores, x86-64 or ARM64	2 GB	200 MB	PWA + approve binary
Server (single)	4 cores	8 GB	20 GB SSD	control plane + SQL small
Server (gamehouse demo)	8+ cores	16–32 GB	100 GB SSD	SQL + media edge + measures
SQL data plane	4+ cores	8 GB min (16 GB recommended)	50 GB SSD	MS SQL 2016–2025

No GPU is required; when a GPU is present the measures use it, otherwise they report `NO DEVICE` and run on CPU.

Software requirements

Component	Requirement
C++ modules	build once with GCC/Clang (Linux) or MinGW/MSVC (Windows); no runtime deps (static)
Data plane	Microsoft SQL Server 2016 / 2017 / 2019 / 2022 / 2025, Express→Enterprise
ODBC	Microsoft ODBC Driver 18 for SQL Server
Web front (optional)	Nginx or Apache or IIS as reverse proxy (URL-rewrite rules below)
PWA	any modern browser; installs to home screen, no store required

OS compatibility matrix

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OS	Version	Status	Notes
RHEL	8 / 9	Supported	primary Linux target
CentOS / Rocky / AlmaLinux	8 / 9	Supported	RHEL-compatible
Fedora	38+	Supported	
Gentoo	current	Supported	source build, `-O2 -march=native`
Debian / Ubuntu	11+ / 22.04+	Supported	build box is Ubuntu 24.04
Windows Server	2016 / 2019 / 2022 / 2025 Datacenter	Supported	detected via RtlGetVersion
Windows	10 / 11	Supported	dev + thin client

The report engine detects the running OS and prints the verdict automatically.

Linux installer (RHEL / CentOS / Gentoo)

```
# 1. build (RHEL/CentOS: dnf install gcc-c++ ; Gentoo: emerge sys-devel/gcc)
g++ -std=c++17 -O2 -march=native ggw_report.cpp -o /usr/local/bin/ggw_report
#    ...repeat for each module, or run build-all.sh

# 2. ISO / drive mount (SCSI and compatible controllers)
#    the installer ISO mounts read-only; sd* = SATA/SCSI/USB, works out of the box:
mount -o ro /dev/sr0 /mnt/iso        # optical / ISO image
mount -o ro /dev/sdb1 /mnt/media     # SCSI / USB media
#    verify the block device first:
lsblk -o NAME,TYPE,SIZE,TRAN         # TRAN shows sata/usb/scsi/nvme

# 3. ODBC for the data plane (RHEL family)
dnf install unixODBC msodbcsql18
```

Windows Server installer

```
1. Install "Microsoft ODBC Driver 18 for SQL Server" (msodbcsql.msi).
2. Copy the pre-built ggw_*.exe (static, no VC++ redist needed) to C:\NetSwitch\.
3. Install SQL Server (Express is fine for the demo) and run ggw_scaling.sql.
4. Open port 1433 (SQL) and your HTTP port in Windows Firewall (below).
```

Ports, sockets & firewall

Service	Port	Proto	Notes
MS SQL	1433	TCP	data plane
HTTP control	8080 (or 80)	TCP	server-cpp / behind proxy
HTTPS	443	TCP	TLS 1.3 (secure module)
SIP	5060	UDP	voice-edge signalling
RTP media	10000–20000	UDP	voice-edge media (range)

Linux firewall (firewalld):

```
firewall-cmd --permanent --add-port=1433/tcp --add-port=443/tcp \
```

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```
                --add-port=5060/udp --add-port=10000-20000/udp
firewall-cmd --reload
```

Windows firewall:

```
netsh advfirewall firewall add rule name="MSSQL" dir=in action=allow protocol=TCP localport=1433
netsh advfirewall firewall add rule name="RTP"   dir=in action=allow protocol=UDP localport=10000-20000
```

Socket automation: the C++ modules open/close their own sockets; nothing to schedule. On Linux you can wrap any module as a `systemd` socket-activated unit if you want the kernel to hold the listening socket and start the service on first connection.

Service packs, patches, SHAs, upgrades

- **SQL Server** follows Microsoft's Cumulative Update (CU) cadence — apply the latest CU for your major version; there are no numbered SP1–SP5 on 2017+, they were replaced by CUs. Track them on Microsoft's SQL Server build list.
- **Our modules** are versioned by git commit; each release zip carries a `SHA256SUMS` file so you can verify integrity:

```
sha256sum -c SHA256SUMS      # Linux
certutil -hashfile ggw_report.exe SHA256   # Windows
```

- Upgrades are drop-in: replace the binary, no migration; the SQL side uses `CREATE OR ALTER` so re-running the script upgrades functions in place.

Hotlink protection — URL rewrite on HTTP_REFERER

Inspect the incoming `Referer` header, allow only your own domain or empty (direct/typed) traffic, and block the rest with `403` (or redirect to a warning image). Rules for all three front ends:

IIS (web.config, official URL Rewrite module)

```
<system.webServer><rewrite><rules>
  <rule name="block-hotlink" stopProcessing="true">
    <match url=".*" />
    <conditions logicalGrouping="MatchAll">
      <add input="{HTTP_REFERER}" pattern="^$" negate="true" />
      <add input="{HTTP_REFERER}" pattern="^https?://(www\.)?yourdomain\.com/" negate="true" />
    </conditions>
    <action type="CustomResponse" statusCode="403" statusReason="Forbidden"
            statusDescription="Hotlinking not allowed" />
  </rule>
</rules></rewrite></system.webServer>
```

Nginx

```
location / {
    valid_referers none blocked yourdomain.com *.yourdomain.com;
    if ($invalid_referer) { return 403; }   # or: rewrite ^ /warning.png break;
}
```

Apache (mod_rewrite)

```
RewriteEngine On
RewriteCond %{HTTP_REFERER} !^$
RewriteCond %{HTTP_REFERER} !^https?://(www\.)?yourdomain\.com/ [NC]
RewriteRule .* - [F]            # F = 403 Forbidden (or: RewriteRule .* /warning.png [L])
```

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Swap `yourdomain.com` for your real domain. “Empty referrer allowed” covers people who type the URL directly or use a bookmark.

On the region / vendor list (Google ARC, SAP EU/US/JP/KR/APAC/AU/IN/TR, LATAM)

That is a very large integration surface. Nothing in the product talks to those networks today, and I won’t invent compatibility we haven’t built and tested. If you tell me which one is actually first on the roadmap (e.g. “SAP EU” or “one Azure region”), I’ll scope a real connector for that one and document it properly rather than list vendors we don’t integrate with yet.

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